CADFEM-Bench v1.0: the final CAD + FEM engineering benchmark

2026-10-13. This release assembles every CAD/FEM source collected in the project into one benchmark for frontier models. **168 tasks** with **506 graded numerical outputs** in 13 task families come from 10 admitted source collections; 73 more collections are held and 8 excluded, each with its blocker. Every reference is source-published (saved solver results, publisher tables, publisher evaluators, closed forms, NAFEMS targets) except the SimJEB peak-stress/verdict outputs, which use the validated SimJEB grader and are labelled *verifier*. No solver run or model call was made for this release.

Tracks. A: Supplied CAD -> FEM: the model receives the part/assembly CAD and the source analysis set-up and must compute the requested results. **B**: Specification/drawing -> CAD -> FEM: the model receives a drawing or written specification, must build the CAD itself and then run the analysis on its own model. **C**: Design under FEM verification: the model must design or modify a part; the benchmark FEM grader decides validity, pass/fail and mass.

Tiers. gold: open CAD or complete specification, fully stated set-up, numerical published or validated reference. **silver**: reference read from a result legend, part of the set-up from the source's boundary-condition figure. **control**: public NAFEMS benchmarks, used to calibrate the harness.

family	track	tier	tasks	graded outputs	physics	reference class
agentfem-bracket	A	gold	32	64	linear static	published_table
cadworld-freecad-fem	A	gold	1	4	linear static	publisher_evaluator
cycloidal-drive-components	A	gold	15	15	linear static	published_raw_field
feenox-valve-transient	A	gold	1	6	multi-material, nonlinear material, transient thermal	published_table
geopinonet-holed-plate	A	gold	16	32	linear static	published_raw_field
morphing-wing-plates	A	gold	8	24	modal	published_table
nafems-calibration	A/B	control	6	6	linear static, plane stress, thermoelastic	benchmark_target
rigging-plate	A/B	silver	6	6	linear static	published_legend
simjeb-analysis	A	gold	38	342	linear static, multi-load-case	published_table, verifier
femcad-hard	B	gold	5	7	Electromagnetics, Heat Transfer, Solid Mechanics	closed_form, published_table
simjeb-design	C	gold	15	0	design optimisation, linear static	SimJEB grader
simjeb-lighten	C	gold	19	0	design optimisation, linear static	SimJEB grader
simjeb-redesign	C	gold	6	0	design optimisation, linear static	SimJEB grader
total			168	506		

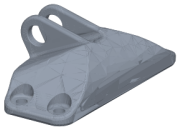
Tasks by track and tier: track A: 111 gold, 3 control, 5 silver; track B: 3 control, 5 gold, 1 silver; track C: 40 gold.

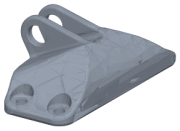
Tolerances. Relative error. Gold defaults 5 % (displacement, frequency, temperature) and 10 % (stress); silver 10 %/20 %. Source tolerances override: CADWorld 5 %, SimJEB 10 % displacement / 15 % stress, FEMCAD-Hard 1 %. Legend peak stresses at load edges (rigging plate) get 20 %; transient temperatures are graded on the rise above the initial state. A task passes only if all its graded outputs pass; results are reported per track and tier and as a family macro-average.

Error policy. Tasks with a known error in their source data were fixed where the fix can be verified and left out otherwise. Fixed: AgentFEM STEP files (metre values under a millimetre header) ship scaled to true millimetres and checked against the FEM nodes; GeoPINONet boundary conditions are stated from the verified fields (the archive's fixed/load STL names are swapped). Left out: DeepWheel (CAD/CSV mass mismatch, unknown density), CADWorld 002 and 003, rigging-plate tests 4 and 5, cycloidal stress outputs (source error estimate 11-49 %) and 13 SimJEB redesign/lighten tasks without a passing human design that proves they can be solved.

Checks at build time. All 239 input files hash-verified; no reference number appears in any prompt or text input (0 leak hits); image inputs were reviewed by eye (problem and boundary-condition figures only). Grader self-test: exact references pass 127/127 numeric tasks, references scaled by 1.3 pass 0. Set-up checks against the source data: cycloidal fixed node sets lie on the stated faces and the per-face loads sum to the deck resultants; GeoPINONet section forces equal -1000 N; the FeenoX valve's Gmsh surface tags equal the OpenCASCADE face indices.

GE jet-engine bracket: analysis of real competition designs (track A) and design, redesign, lighten (track C)





Ti-6Al-4V bracket, four GE load cases at the clevis pin (RBE3), rigid fully fixed bolts. Analysis tasks ask for the per-case maximum displacement (graded against SimJEB's published OptiStruct magnitudes, 10 %), the peak von Mises stress outside 2 mm of the bolt/pin walls (verifier, 15 %) and the pass verdict. Track C: 15 design tasks (GE specification and material/load variants), 6 redesigns of failing designs (mass cap 1.25x) and 19 lightening tasks (-15 % mass), each with a passing human design that proves it can be solved; graded by scripts/simjeb_bench.py (geometry rules, FEM, mass) and placed against the 381 human designs.

Tasks: 78. **Licence:** Dataverse CC0 / README ODC-By; GE bracket CAD from GrabCAD (non-commercial). Local research use. **Source:** SimJEB: simulated jet engine bracket dataset (Whalen, Beyene, Mueller, CGF 2021) / GE bracket challenge.

Example A-SIMJEB-analysis-0 (hidden references):

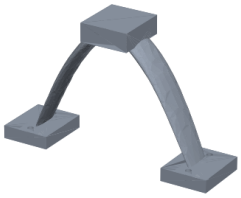
output	unit	reference	tolerance	class	provenance
vertical_max_displacement_mm	mm	0.321826	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_ver_magdisp (OptiStruct)
vertical_peak_von_mises_MPa	MPa	402.833	rel 0.15	verifier	SimJEB-Bench FEM grader (scripts/simjeb_tasks.py)
horizontal_max_displacement_mm	mm	0.239873	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_hor_magdisp (OptiStruct)
horizontal_peak_von_mises_MPa	MPa	266.263	rel 0.15	verifier	SimJEB-Bench FEM grader (scripts/simjeb_tasks.py)
diagonal_max_displacement_mm	mm	0.165667	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_dia_magdisp (OptiStruct)
diagonal_peak_von_mises_MPa	MPa	202.392	rel 0.15	verifier	SimJEB-Bench FEM grader (scripts/simjeb_tasks.py)
torsion_max_displacement_mm	mm	0.174951	rel 0.1	published_table	SimJEB metadata CSV id 0 column max_tor_magdisp (OptiStruct)
torsion_peak_von_mises_MPa	MPa	274.958	rel 0.15	verifier	SimJEB-Bench FEM grader (scripts/simjeb_tasks.py)
passes	bool	True	bool 0	verifier	SimJEB-Bench FEM grader verdict

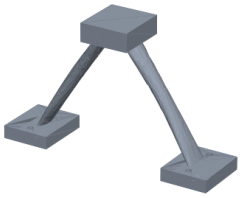
Example prompt (A-SIMJEB-analysis-0), first lines:

```
Design task: the GE jet engine bracket. Units: millimetres, newtons, MPa. Material: Ti-6Al-4V (E = 113.8 GPa, nu
= 0.342, density 4.43 g/cm^3, yield strength 903 MPa). The bracket is bolted to a flat mounting surface, the
plane z = 0, with nothing below it, and carries a clevis for a pin.
Interfaces: (a) four vertical (+z) bolt holes of diameter 10.3124 mm (0.406 in) at (x, y) = (0.0000, 0.0000);
(52.0472, 1.5805); (0.0000, -148.1248); (38.0833, -146.9684); each hole wall must start within 1.0 mm of z = 0
and be at least 5.5 mm long; the bolts are rigid and fully fixed. (b) a clevis of two lugs, each with a bore of
diameter 19.1008 mm on the pin axis through (-20.9236, -74.6978, 44.7246) with direction (-0.030352, 0.999539,
0); measured along the axis from that point, the lug bores span 11.112 to 17.463 mm and -17.463 to -11.113 mm;
the gap between the lugs must stay empty.
Design envelope: x -41 to 72, y -168 to 22, z 0 to 65 mm. Minimum wall / feature thickness 1.27 mm (at most 2 % of
the surface may be thinner). One solid.
Loads, four separate linear static cases, each acting at the point (-21.0359, -75.0647, 44.8010) and spread ...
```

Flags: peak stress and verdict use the validated benchmark grader (no published equivalent)

AgentFEM double-arm support brackets (publisher test split)





Aluminium bracket (E 70 GPa), foot undersides fixed, uniform downward seat traction of 1000 or 10000 N. Outputs: mean seat deflection and maximum displacement (publisher summary, 10 %; publisher mesh not converged). The source STEP declared mm but held metre values; the shipped copy is scaled to true mm and checked against the FEM nodes.

Tasks: 32. **Licence:** CC BY 4.0. **Source:** AgentFEM Bracket Elasticity (Haoming Luo, 2026).

Example A-AGENTFEM-fresh_test_000 (hidden references):

output	unit	reference	tolerance	class	provenance
seat_mean_displacement_mm	mm	0.0983548	rel 0.1	published_table	publisher cases/fresh_test_000/summary.json
max_displacement_mm	mm	0.175628	rel 0.1	published_table	publisher cases/fresh_test_000/summary.json

Example prompt (A-AGENTFEM-fresh_test_000), first lines:

```
Support bracket (one member of a parametric double-arm bracket family). geometry.step holds one solid in
millimetres (height 189 mm).

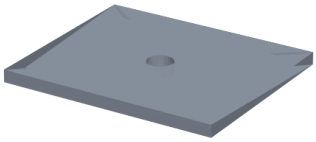
Material: linear isotropic elastic, E = 70 GPa, Poisson ratio 0.33 (density 2700 kg/m^3, gravity not applied).
Supports: the foot undersides, i.e. every face lying in the plane z = 0, are fully fixed (all displacement
components zero).
Load: a uniform downward traction of 1.98728 MPa in the -z direction over the top seat face (the face in the
plane z = 189 mm); total 10000 N.
Analysis: 3-D linear elasticity, small displacements. The publisher used quadratic tetrahedra with a nominal 9 mm
mesh.

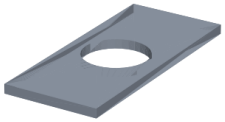
Outputs:
- seat_mean_displacement_mm [mm]: mean of the downward displacement -u_z over the top seat face (z = 189 mm)
- max_displacement_mm [mm]: maximum displacement magnitude anywhere in the bracket

Write results.json as {"outputs": {"seat_mean_displacement_mm": <number>, "max_displacement_mm": <number>}} with
the units listed. ...
```

Flags: source STEP unit error fixed: shipped copy scaled x1000 to true millimetres and checked against the FEM nodes; one parametric family: report as a family macro-average

GeoPINONet holed steel plates (validation split)





Structural steel, one end fixed, 1000 N on the opposite end face: compression (-y) and lateral (+x). Outputs: maximum displacement per case from the published ANSYS nodal CSVs (5 %). The archive's fixed/load STL names are swapped relative to the fields; supports and loads are therefore stated by geometry.

Tasks: 16. **Licence:** CC BY 4.0. **Source:** GeoPINONet holed-plate FEM dataset (Urrutia Valdes, La Fe-Perdomo, Rodriguez Soto).

Example A-GEOPINO-PLATE-1 (hidden references):

output	unit	reference	tolerance	class	provenance
max_displacement_compression_mm	mm	0.000439557	rel 0.05	published_raw_field	max |u| over the published ANSYS nodal CSV (1_compression.csv / _lateral_bending.csv)
max_displacement_lateral_mm	mm	0.00303832	rel 0.05	published_raw_field	max |u| over the published ANSYS nodal CSV (1_compression.csv / _lateral_bending.csv)

Example prompt (A-GEOPINO-PLATE-1), first lines:

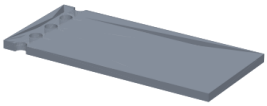
```
Holed steel plate. part.step is one solid in millimetres: a rectangular plate 176.914 mm (x) by 148.475 mm (y) by
10.000 mm (z) with a through hole along z.
Material: ANSYS default Structural Steel, linear isotropic elastic, E = 200 GPa, Poisson ratio 0.3. Linear
static, small displacements, no gravity.
Supports: the end face at y = -74.2373 mm is fully fixed (all displacement components zero).
Two separate load cases, each a total force of 1000 N applied as a uniform traction over the opposite end face (y
= 74.2373 mm):
(1) compression: 1000 N along -y (towards the fixed end);
(2) lateral bending: 1000 N along +x (in-plane shear of the free end).

Outputs:
- max_displacement_compression_mm [mm]: maximum displacement magnitude, load case 1
- max_displacement_lateral_mm [mm]: maximum displacement magnitude, load case 2

Write results.json as {"outputs": {"max_displacement_compression_mm": <number>, "max_displacement_lateral_mm":
<number>}} with the units listed. ...
```

Flags: source errors not carried over: BCs stated from the verified fields (archive STL names are swapped), thickness from the CAD (README says 20 mm, CAD and FEM 10 mm)

Active-camber morphing-wing cantilever plates (modal)





FDM-printed PLA plates, uniform and with a 1 mm flexible strip, clamped at the end of a 19 mm grip. Outputs: first bending, first torsion and second bending frequencies from the publisher workbook (Abaqus thick shell / Nastran plate models, 5 %).

Tasks: 8. **Licence:** CC BY 4.0. **Source:** Experimental and numerical data of candidate structural models for active camber morphing wings (Huang & Zhang).

Example A-MORPHWING-Test1 (hidden references):

output	unit	reference	tolerance	class	provenance
f_1B_Hz	Hz	109.05	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S4 (Abaqus thick shell S8R)
f_1T_Hz	Hz	416.67	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S5 (Abaqus thick shell S8R)
f_2B_Hz	Hz	672.94	rel 0.05	published_table	'1-1 Modal Test Experiment and Numerical Results.xlsx' cell S6 (Abaqus thick shell S8R)

Example prompt (A-MORPHWING-Test1), first lines:

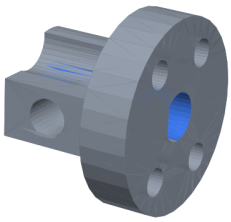
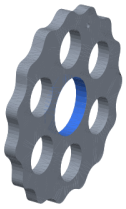
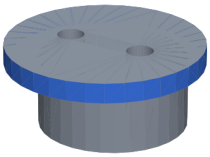
```
Cantilever plate specimen for an active-camber morphing-wing study (FDM-printed PLA): a uniform 4 mm plate.
part.step is one solid in millimetres, 109 x 50 x 4 mm, with a 19 mm clamping grip at x = 0-19 mm (the scalloped
clamp region).
Model the vibrating structure only: the free part x >= 19 mm, clamped (all displacements and rotations zero)
along its whole cross-section at x = 19 mm. The grip is held in the fixture and is not part of the model.
Material: PLA treated as isotropic linear elastic, E = 2107.5 MPa, Poisson ratio 0.3, density 1180 kg/m^3.
Analysis: undamped natural frequencies (modal analysis), no prestress, no gravity. Identify the modes by their
shapes.

Outputs:
- f_1B_Hz [Hz]: first out-of-plane bending mode
- f_1T_Hz [Hz]: first torsion mode about the span (x) axis
- f_2B_Hz [Hz]: second out-of-plane bending mode

Write results.json as {"outputs": {"f_1B_Hz": <number>, "f_1T_Hz": <number>, "f_2B_Hz": <number>}} with the units
listed. ...
```

Flags: reference is a shell-model solution of the free length; solid models converge to it within about 1-2 %

3D-printable cycloidal gearbox: component static analyses

FreeCAD/CalculiX studies of 13 printed PETG-CF parts and 2 steel pins. Blue: fixed faces; red: loaded faces (uniform traction, resultant from the source-corrected deck). Output: maximum displacement (5 %) read from the saved FRD files. Stress is not graded: the source's own CalculiX error estimate at the stress peaks is 11-49 %.

Tasks: 15. **Licence:** No repository license; FCStd metadata 'All rights reserved'. Local research only. **Source:** 3D-printable cycloidal drive FEA (jagathsrujan/3d-printable-cycloidal-drive).

Example A-CYC-Coupler-PETGCF (hidden references):

output	unit	reference	tolerance	class	provenance
max_displacement_mm	mm	0.00761747	rel 0.05	published_raw_field	fem/petg_cf/FEM_Coupler_PETGCF/Coupler_fixed.frd (saved CalculiX FRD of the source-corrected deck)

Example prompt (A-CYC-Coupler-PETGCF), first lines:

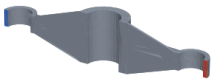
```
Component analysis: the output coupler of an open-source, 3D-printable 15:1 cycloidal gearbox. part.step holds
the single solid in millimetres (33 faces, volume 2063 mm^3). faces.json lists every face with its index, type,
area, centroid and normal/axis; use the geometry, not a CAD system's own face numbering.

Material: PETG-CF, linear isotropic elastic, E = 3000 MPa, Poisson ratio 0.36.
Analysis: linear static, small displacements, no gravity, no contact.
Supports: all three displacement components are zero on face 5: cylinder, area 33.52 mm^2, centroid (76.619,
-0.908, 22.737) mm, radius 2.5 mm, axis (1.0, 0.0, -0.0) through (79.025, -0.0, 20.5) (hole/bore surface); face
6: cylinder, area 117.81 mm^2, centroid (76.525, 1.588, 20.397) mm, radius 2.5 mm, axis (1.0, 0.0, -0.0) through
(79.025, -0.0, 20.5) (hole/bore surface).
Load: total force (100.052, 0.000, 0.000) N applied as a uniform surface traction (constant force per unit area)
over the loaded face(s): face 14 carries (100.052, 0.000, 0.000) N.
For reference, the source solved this with second-order tetrahedra (Netgen, maximum element size 2 mm).

Outputs:
- max_displacemen ...
```

Flags: source author corrected FreeCAD CLOAD units by 1000x; loads stated from the corrected deck

CADWorld FreeCAD FEM task



Publisher geometry, material, constraints and mesh protocol (Gmsh, first-order tetrahedra); references are the values the publisher's evaluator checks (5 %). 002 (edge support, auxiliary load object) and 003 (20 MN on a 30 mm specimen, 811 mm displacement) are left out.

Tasks: 1. **Licence:** Apache-2.0 (publisher HF metadata). **Source:** CADWorld FEM benchmark (Zdong104/CADWORLD; HF Zihan1004/CADWORLD).

Example A-CADWORLD-001 (hidden references):

output	unit	reference	tolerance	class	provenance
max_displacement_mm	mm	31.145	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Displacement Magnitude' ma
max_von_mises_MPa	MPa	705.453	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'von Mises Stress' max (pub
max_tresca_MPa	MPa	397.017	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Tresca Stress' max (publis
max_principal_MPa	MPa	863.519	rel 0.05	publisher_evaluator	CADWorld reference CSV evaluation_examples/fixtures/fem/freecad-fem-001.csv column 'Major Principal Stress' ma

Example prompt (A-CADWORLD-001), first lines:

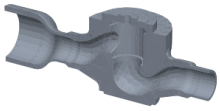
```
FreeCAD/CalculiX benchmark part (Multi-feature aluminium part, end load). part.step is the single solid in
millimetres (38 faces, bounding box [-130.0, -0.0, -30.0, 130.0, 30.0, 30.0] mm). faces.json describes every
face.

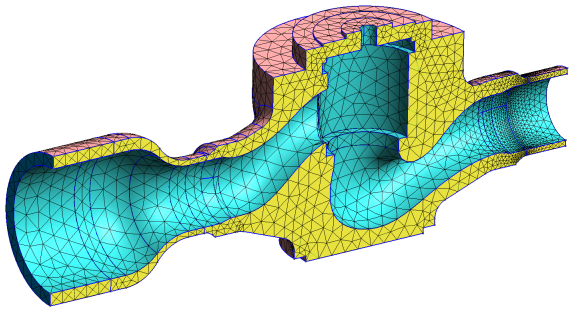
Material: AlMg3F24, linear isotropic elastic, E = 70000 MPa, Poisson ratio 0.3.
Analysis: linear static, small displacements, no gravity.
Supports: fixed (all displacement components zero) on face 6: plane, area 100.00 mm^2, centroid (-127.5, 0.0,
0.0) mm, outward normal (-0.0, -1.0, -0.0).
Load: total force 200 N along (0, 1, 0) applied uniformly over face 38: plane, area 100.00 mm^2, centroid (127.5,
-0.0, 0.0) mm, outward normal (-0.0, -1.0, -0.0).
Mesh protocol of the publisher's reference solution (results are compared under this protocol): Gmsh 3-D mesh,
first-order (linear) tetrahedra, maximum element size 2.0 mm, minimum 0.5 mm, CalculiX solver. Report nodal
extrema of that solution.

Outputs:
- max_displacement_mm [mm]: maximum nodal displacement magnitude
- max_von_mises_MPa [MPa]: maximum nodal von Mises stress
- max_tresca_MPa [MPa]: maximum nodal Tresca stress
- max_principal_MPa [MPa]: maximum nodal ...
```

Flags: protocol-bound: publisher mesh (first-order tets) fixes the reference; publisher tolerance 5 %; linear-elastic stresses exceed material yield; computational benchmark, not a strength verdict

Power-plant valve: nonlinear transient heat conduction





Carbon-steel nozzle and stainless bodies, ASME temperature-dependent properties, internal temperature ramp 40-250-40 C over 1000 s, weak external convection. Outputs: through-wall temperature differences at two stress classification lines (heat-up and cool-down extrema, 10 %) and outer-wall temperatures at 100 s, from the published FeenoX history.

Tasks: 1. **Licence:** GPL-3.0. **Source:** FeenoX tutorial #3: heat conduction (valve with carbon and stainless steel parts).

Example A-VALVE-transient (hidden references):

output	unit	reference	tolerance	class	provenance
scl2_dT_max_K	K	14.029	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl2_dT_min_K	K	-15.183	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_dT_max_K	K	26.128	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_dT_min_K	K	-28.159	rel 0.1	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl2_T_outer_100s_C	C	235.837	rel_offset 0.05	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)
scl4_T_outer_100s_C	C	223.738	rel_offset 0.05	published_table	published FeenoX history valve-1.csv (192 adaptive time samples)

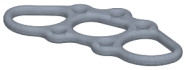
Example prompt (A-VALVE-transient), first lines:

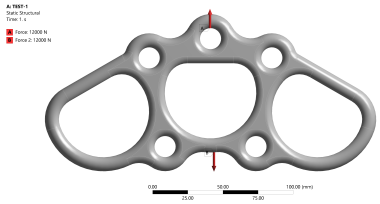
```
Nonlinear transient heat conduction in a power-plant valve: a carbon-steel nozzle welded to stainless-steel valve
bodies (half model, symmetry plane y = 0). valve.brep holds 4 solids in millimetres (112 faces). Solid 2 (the
nozzle, x from 0 to 65 mm) is carbon steel; solids 1, 3 and 4 are stainless steel SA-312. faces.json lists every
face with its index and its boundary group; valve-surfaces.png and valve-materials.png are the source's figures.

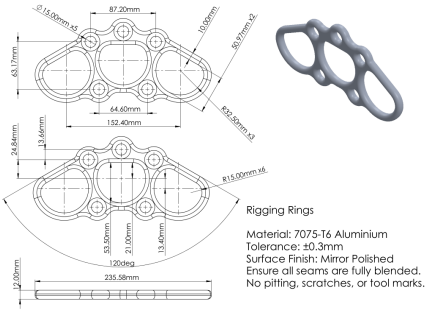
Materials: temperature-dependent conductivity k(T) and thermal diffusivity kappa(T) from asme-properties.fee
(ASME II Part D tables, T in deg C): carbon steel uses k_carbon and kappa_carbon, stainless uses k_312 and
kappa_312. As in the source, k from the tables is in W/(m K) and is multiplied by 1e-3 to W/(mm K); kappa is used
as tabulated (mm^2/s). The volumetric heat capacity follows as k/kappa.
Boundary conditions: 'internal' faces: prescribed temperature T = Tint(t) from internal-temperature.csv (linear
interpolation: 40 C at t = 0, 250 C from 100 s to 500 s, back to 40 C at 600 s, held to 1000 s). 'external' faces:
convection h = 1e-6 W/(mm^2 K) to 50 C. 'symmetry' and both end faces: ...
```

Flags: through-wall differences use the published 192-sample history; 10 % tolerance on differences; temperatures at 100 s graded on the rise above the 40 C initial state

7075-T6 rigging plate: five ANSYS load tests and the drawing-to-CAD variant (silver)







Supplied STEP (track A) or the dimensioned manufacturing drawing (track B) plus the source's boundary-condition figure per test. Output: maximum von Mises stress from the original legend (20 %). Track B also checks the built part's volume and extents against the source CAD (2 %). Tests 4 and 5 are left out (conflicting load labels; revised TEST-5R results).

Tasks: 6. **Licence:** CERN-OHL-S-2.0. **Source:** OpenRiggingPlate (mpangch).

Example A-RIGGING-test1 (hidden references):

output	unit	reference	tolerance	class	provenance
max_von_mises_MPa	MPa	526.74	rel 0.2	published_legend	FEA Analysis/Test 1 (2697.7 lbf)/Stress.png legend '526.74 Max'

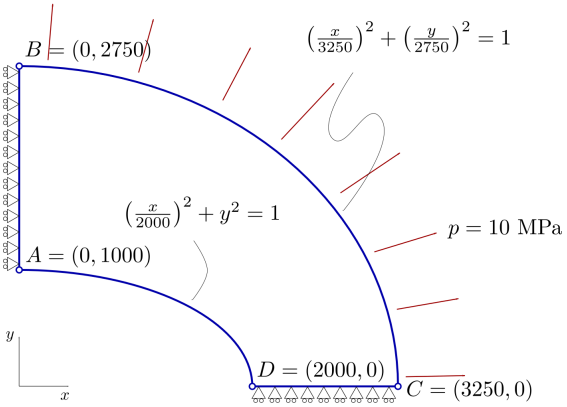
Example prompt (A-RIGGING-test1), first lines:

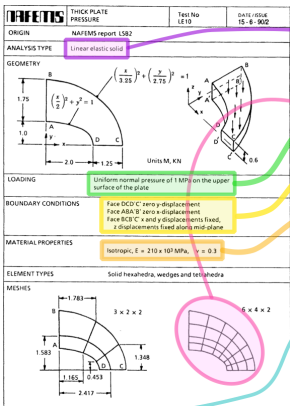
```
Hard-anodised 7075-T6 rigging plate, load test 1 of the source's seven ANSYS load cases (Test 1 (2697.7 lbf)).
riggingplate.step is the single solid in millimetres (235.58 x 111.90 x 12 mm, 145 faces).
Material: aluminium 7075-T6, linear isotropic elastic (standard handbook values; the source did not publish its
card, and the graded stress does not depend on Young's modulus for these loads).
Analysis: linear static, small displacements (ANSYS Mechanical in the source).
Boundary conditions: exactly as shown in boundary-conditions.png (the source's own ANSYS figure): each labelled
scope (A, B, ...) is a face or set of faces of the plate; 'Force' applies the stated total force uniformly
distributed over the scoped faces in the drawn direction; 'Remote Displacement' fixes the scoped faces through a
rigid remote point (all displacements and rotations zero). Where only two equal and opposite forces are applied
and no support is drawn, the plate is self-equilibrated; remove rigid-body motion without adding load (e.g. weak
springs or inertia relief), as the source did.
Legend of this test's figure: A Force: 12000 N; B; Force 2: 12000 N.

Ou ...
```

Flags: load scopes given by the source figure, not by face IDs; reference is a rounded legend maximum; peak at load-face edges is mesh-sensitive (20 % tolerance)

Calibration and closed-form tasks





NAFEMS LE1 (elliptic membrane), LE10 (thick elliptic plate) and LE11 (thermoelastic cylinder/taper/sphere), each as track A (supplied geometry) and track B (build CAD from the published figure). References are the FeenoX outputs, which equal the NAFEMS targets (5 %). Control tier: answers may be recalled.

FEMCAD-Hard (track B, gold): build the part from the written specification with CadQuery, analyse it with an open-source FEM code, report converged values. References are closed forms or values stated on the Wolfram FEM documentation pages, with the task's own 1 % tolerance.

task	title	outputs (hidden reference, tolerance)
B-FEMCAD-031	Spherical capacitor shell	capacitance_pF = 1.85972 (0.01, closed_form)
B-FEMCAD-057	L-bracket with a filleted inner corner	max_von_mises_MPa = 230.197 (0.05, published_table)
B-FEMCAD-058	Spherical pressure hull (bathyscaphe) under sea pressure	u_outer_nm = -123.29 (0.01, closed_form); u_inner_nm = -144.149 (0.01, closed_form); max_von_mises_MPa = 0.307377 (0.01, closed_form)
B-FEMCAD-062	Plate with a hole in tension	max_von_mises_MPa = 14.9 (0.05, published_table)
B-FEMCAD-067	Thick-walled tube heated on an inner band	T_probe_K = 332.97 (0.01, published_table)

Source registry: held and excluded collections

73 held collections contain real CAD/FEM data with a specific blocker; 8 are excluded. Held blockers: CAD only in a proprietary native format (19); supports/loads/materials not recoverable as CAD-face set-up (19); conflicting or unmatched source references (18); no deterministic, mesh-independent numerical reference (8); no B-rep CAD (mesh, scan or paper geometry only) (5); needs a proprietary solver/material model to define the problem (4). Largest pools that could be admitted with further work: Code_Aster validation cases (50 cases, 761 published checks), SFEM (61), GeoPINONet lugs (80), Bath auxetics (18). Tasks left out of admitted sources because of source errors are listed under each source's decision.

collection	status	candidates	blocker / decision
SimJEB GE jet engine bracket (381 human designs)	admitted (78 tasks)		Analysis displacement vs SimJEB published OptiStruct; peak stress/verdict and design tasks via the validated benchmark FEM grader. 13 redesign/lighten tasks without a passing human witness design (unproven feasibility) are left out.
AgentFEM bracket elasticity (176 pairs)	admitted (32 tasks)		Publisher test split (32) admitted with the source STEP unit error fixed (metre-valued coordinates under a mm header; shipped copies scaled to true mm and checked against the FEM nodes); train/validation (144) kept out to avoid redundancy within one parametric family.
GeoPINONet holed plates (80 pairs)	admitted (16 tasks)		Validation split (16) admitted with BCs stated by geometry (archive fixed/load STL names are swapped relative to the fields).
Morphing-wing cantilever plates (modal)	admitted (8 tasks)		Tests 1-3, 5-9 admitted; Test 4 excluded (STEP fef50 % vs result block f75 %).
3D-printable cycloidal drive component FEM (31 saved solutions)	admitted (15 tasks)		15 admitted (PETG-CF set + steel pins), displacement only: the source's CalculiX error estimate at the stress peaks is 11-49 %, so stress is not graded. PLA reruns of the same geometry/loads, the axial-cam variant, the mesh-only housing FRD and the uncorrected disk deck are not separate tasks.
CADWorld FreeCAD/CalculiX FEM tasks (3)	admitted (1 tasks)		001 admitted; 002 left out (support on a single edge, load on an auxiliary Box object; 76.6 GPa peak is a singular-support artefact); 003 left out (20 MN on a 30 mm specimen gives 811 mm displacement, outside small-displacement analysis).
NAFEMS LE1/LE10/LE11 via FeenoX	admitted (6 tasks)		Calibration tier (public benchmarks, answers may be recalled); GNL5 large-deflection case not packaged.
FEMCAD-Hard v1 (Wolfram FEM docs)	admitted (5 tasks)		Five tasks with closed-form/page-stated references admitted.
FeenoX power-plant valve transient thermal	admitted (1 tasks)		Admitted; Gmsh surface tags verified equal to OpenCASCADE face indices.
OpenRiggingPlate 7075-T6 (7 ANSYS load tests + drawing)	admitted (6 tasks)		Silver: references are rounded legend maxima; load scopes come from the source figures. Tests 4 (README/folder/figure load labels disagree) and 5 (results labelled TEST-5R, a revised set-up) left out.
DeepWheel free-free wheel modes (10 downloaded of 904)	held	10	CAD mass at the publisher density (2680 kg/m^3) is 3.25 % below the publisher CSV mass; the density behind the published frequencies cannot be determined.
SFEM stochastic linear elasticity (61)	held	61	Fields are medians over random, unseeded load clumps whose locations were not stored: no deterministic set-up to give a model.
Bath curved-link auxetic lattices (18)	held	18	CAD is Autodesk Inventor .ipt only; the ANSYS .dat decks carry the mesh, so no open CAD exists for a CAD->FEM task.
Auxetic meta-shells torsional buckling (6)	held	6	References are torque-twist curves of nonlinear Abaqus shell models; the shell section, contact and loading must be transcribed from the decks first.
Scan-based cold-formed steel columns (3)	held	3	Geometry is a scan STL only (no B-rep); imperfections are encoded in the mesh.
NAFEMS GNL5 large-deflection beam	held	1	Not in the pilot packaging; geometric-nonlinear prompt and load history not yet transcribed.
Code_Aster validation cases (50, 761 published checks)	held	50	No CAD (GIBI/MED meshes; 28 geometry scripts); checks are keyed to mesh groups. Prompts must be transcribed from the French V-documents.
FEMCAD Source v2 (Wolfram, 100 linked cases)	held	100	Task specifications only: no gold answers were extracted (no solver runs by policy).
COMSOL application-gallery joint tasks (50)	held	50	No numerical references or tolerances; source models need COMSOL.
DeepJEB brackets (2,138 scalar results)	held	2138	Only one STEP downloaded (shell, not solid); same GE load set as SimJEB, which is admitted.
COMSOL legacy studies (turbine stator, gears, muffler, ...)	excluded	7	COMSOL-only solved models; quantitative fields require COMSOL export.

collection	status	candidates	blocker / decision
FrontISTR FreeCAD benchmarks: gear, can, bike frame, spring	held	4	Only peak von Mises is published and it keeps rising with refinement (gear 334 -> 397 -> 442 MPa): no mesh-independent target.
GeoPINONet 3-D lugs (80)	held	80	Pin-bore load distribution (uniform force vs bearing) is not documented, and the archive's fixed/load STL names are swapped.
UCSM sheet-metal inverse forming (8)	held	8	LS-DYNA inverse one-step strains on surface CAD; material card and process input not supplied.
Tricuspid heart valve (Abaqus/Explicit VUMAT)	held	1	Shell-surface IGES and a user material subroutine; replay needs the published VUMAT inside Abaqus.
fcFEM collapse examples (5)	held	5	Saved snapshots cannot be matched to load factors/increments from the published parameter files.
Aerospace wing box (IGES + APDL)	held	1	IGES faces only; geometry/unit/set-up correspondence with the published results unresolved.
Cemented hip implant contact (Leeds)	held	1	Pelvis/cement are open IGES surfaces; only one of four wear states maps to an executable deck.
HIRENASD wing modes (NASA)	held	1	Mode-shape fields only; frequency file not public; frame correspondence unresolved.
Blade dovetail reduced model (PoliTo)	held	1	Reduced mass/stiffness matrices only, no response results.
Fokker F28 fuselage crash (TU Delft)	held	3	Results only as plots mixing FEM and experiment; no machine-readable response.
WAAM wall thermal history and distortion (AIMEN/SCAI)	held	1	No solver deck (moving heat source parameters only in PDFs); VMAP unit tags undefined.
Capriccio FE/MD fracture, Digimat cure, bimetal strip, SOFA actuator, tapered spring	held	6	Coupled FE/MD or proprietary-material (Digimat/Abaqus) set-ups; no open CAD; fields partly non-finite.
FreeCAD Code_Aster laminate + CalculiX comparison; Hannover milling	held	3	Laminate result units and touched solver objects unresolved; milling has measurements, not FEM.
Test-bench frame (FreeCAD/CalculiX)	held	1	Displacement is prescribed; the only response target is a coarse-mesh peak stress.
ACP composite kiteboard (Ansys tutorial)	held	1	Layup lives in ACP; 'SOLVED' archive lacks results; reference from a different software version.
Ansys ACP workshops (4)	excluded	4	No saved results.
PrePoMax tutorial geometries (12)	held	12	Idealised tutorial parts with plot-only results in a manual.
Modular helical marine anchor (OpenEI, 3 load sets)	held	3	CAD only as DesignModeler/SolidWorks/Parasolid; analysis uses idealised equivalent members.
PyDPF composites examples (5)	held	5	Native SpaceClaim geometry, ACP layups; parser failures on two projects.
Layered composite pipe (2 revisions)	held	1	Two conflicting saved revisions; orthotropic shell layup; IGES is a surface.
Pneumatic multi-link gripper (nonlinear contact)	held	3	Joint/contact definitions not recoverable without the 2026R1 archive decoder; baseline/A1 have tables only.
Fin Ray 3-finger hyperelastic contact	held	1	Full three-finger CAD not published (one finger STEP + assembled mesh).
Bolted nozzle-flange sector (3 repos)	held	4	DesignModeler-only CAD; README vs deck conflicts (units, materials, temperatures).
Formula SAE chassis (4 static cases)	held	4	SpaceClaim-only beam CAD; decoded fields conflict with the published displacements.
Go-kart chassis modal/harmonic	held	1	SolidWorks-only CAD; plot/report references.
IIT Mandi shear frame (4 modal variants)	held	4	Abaqus CAE-only CAD; joint modelling and damage-case revision mapping unresolved.
Double-wishbone suspension	held	1	Supplied CAD revision cannot be matched to the baseline or redesigned stress state.
Perforated brake disc steady thermal	held	1	Peak is a prescribed boundary value; convection/temperature face scopes must be mapped from the deck.
Horizontal pressure vessel (Nimonic 75)	held	1	IGES surfaces only; saddle support scope undocumented.
Pneumatic cylinder cover	held	1	Source support is a single fixed node plus frictionless node sets; not expressible as CAD-face BCs.
IC engine connecting rod (2 saved cases)	held	2	Traction/support scopes must be mapped from the ANSYS deck node components to STEP faces.
Connecting rod (SolidWorks/ANSYS report)	held	1	STEP in inches with a 4.98 m extent; plots in CGS; support scope undocumented.
Twin-wheel nose landing gear	held	1	Only a minimum safety factor is published; load/support scopes only in report figures; materials partly unconfirmed.

collection	status	candidates	blocker / decision
Single-cylinder crank assembly	held	1	Material cards undocumented; thermal references conflict.
Ventilated brake rotor (thermal + structural)	held	1	IGES faces with no solid; ten fixed faces and pressure scopes not identified.
Pipe-flange assembly (2 connection models)	held	2	SolidWorks-only CAD; load scope undocumented; drawing inch/mm conflict.
Bolted shelf with pretension and contact	held	1	SpaceClaim-only CAD; saved results disagree with the report (0.028 vs 0.025 mm).
CubeSat frame archive	held	1	Archive geometry does not match the published CubeSat.
Elmer conjugate cooling (radiator, tube)	held	2	Set-up lives in native FreeCAD/Elmer objects; README vs native heat inputs differ; fields undershoot ambient.
Titanium wing static + prestressed modal	held	2	DesignModeler-only CAD.
Wheel rim static + modal (5 analyses)	held	5	SolidWorks/SpaceClaim-only CAD.
Quadcopter frame static/modal/harmonic	held	3	Native assemblies only; harmonic load cards conflict with the report; modal uses point masses.
Curved steam pipe and elbow thermal	held	4	SpaceClaim-only CAD; problem figures not complete drawings.
Seven-fin heat sink steady thermal	held	1	Convection scope (41 of 55 faces) not identifiable; the published temperatures imply ~0.047 m^2 of convective area, less than the fin sides alone (0.054 m^2).
NX fin-array heat sink	held	1	NX-only CAD; heat input and film coefficients absent.
Spur gear pair	held	1	Load, material and contact law not specified.
Spring shock absorber	held	1	Loads and supports not documented.
Screw jack	held	1	README and plot references conflict; native CAD only.
Knuckle joint	held	1	Inventor-only CAD; STEP not uploaded.
Multi-leaf spring modal	held	1	Two conflicting frequency sets; surface-only IGES.
Steel dogbone static/buckling control	held	2	Discovery-only CAD; supports undocumented.
Piston-pin-connecting rod	held	1	Part files missing from the assembly; README conflicts.
Piston assembly (duplicate source)	excluded	0	Byte-identical duplicate of the piston source.
Piston thermal-barrier coating	held	3	Export/report revision mapping unresolved; obsolete result state.
Heat-assisted incremental forming	held	6	Partial CAM CAD only; conflicting job tables.
Kresling origami	held	1	Inventor/Parasolid-only CAD; curve-to-case mapping unresolved.
Bolted plywood truss	held	1	Idealised 2-D truss export with unknown units.
Haunched steel portal frame	held	4	Native CAD only; inch/mm drawing conflict; load scope ambiguous.
Aquantis tidal turbine	held	1	CAD/FEM pairing not established (45 m CAD vs 41.5 m FEM).
Splined differential shaft	held	1	Report displacement exceeds the part length ~963x; conflicting reports.
Carbon-fibre UAV frame (62 bodies)	held	1	Orthotropic properties and 132 bonded contacts not frozen; report-only references.
External gear-pump housing	held	1	SolidWorks-only CAD; configuration/plot pairing unknown.
Ribbed mounting flange (drawing + 55-row table)	held	1	Drawing dimension conflicts; screenshot values match no table row; supports undocumented.
Paper drawings with FEM (pump shaft, ...)	held	10	Paper-only geometry and results; solver set-ups incomplete or inconsistent.
6-axis robot arm	excluded	1	Rigid-body joint motion only; no deformable FEM.
CAD/drawing leads without FEM output	excluded	7	No saved or published FEM result.
Test-bench strip restarts	excluded	4	Duplicate frame CAD; strip results have no matching CAD.
TU Wien e-bike frame	held	1	SolidWorks-only CAD; plot-only, incomplete solver specification.
SFEM method re-run on ABC CAD	excluded	489	Loads generated locally; the user asked for real engineering set-ups, not invented loads.
Engineering FEM v1	excluded	14	Superseded by later collections.

Files: candidate package (give to the model) cadfem-bench-v1-candidate.zip; evaluator package (keep hidden) cadfem-bench-v1-evaluator.zip with references.jsonl, registry.jsonl, score.py, validation.json and the dataset card; builder scripts in cvpr2027/scripts/cadfem_bench/. Licences differ per source and several sources are local-research-only: do not publish the candidate package as a whole. No model results yet.